EduChain

Soulbound Academic Credential Registry on BOT Chain

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PROTOCOL OVERVIEW

EduChain is a soulbound certificate registry deployed on BOT Chain that enables accredited educational institutions to issue, manage, and revoke tamper-proof academic credentials on-chain. Certificates are permanently bound to a student wallet address and cannot be transferred, while a hash/CID of the actual credential file is stored on-chain for off-chain retrieval. Anyone can verify a certificate by its ID without a wallet, making credential verification universally accessible.

A technical overview of the protocol, its design, core functions, security properties, and roadmap.

1. Problem Statement

Academic credential fraud costs institutions, employers, and governments billions annually. Current approaches are fragmented:

- Paper certificates — easily forged, destroyed, or lost.
- Centralised registries — require institutional API access; vulnerable to data breaches.
- Third-party verification services — add cost, latency, and intermediary trust requirements.
- No portable standard — employers must contact each institution individually.

2. Solution

EduChain encodes the full credential lifecycle on-chain. An admin governs which institutions may issue certificates. Institutions issue soulbound certificates that are permanently tied to a student's address. Revocation is possible by the issuing institution. Public verification requires no wallet.

2.1 Roles

`Role`
Permissions

Admin

Register / remove institutions; transfer admin role

Institution

Issue certificates to students; revoke own certificates

Student

View own certificates; share verify links

Public

Verify any certificate by ID — no wallet required

3. Contract Architecture

3.1 Data Model

Institution: name (string), active (bool).

Cert: issuer address, student address, studentName, title, fileHash (IPFS CID or SHA-256), issuedAt timestamp, revoked flag.

3.2 Soulbound Guarantee

Certificates are stored in a private array and exposed only through view functions. There are no transfer, approve, or setApprovalForAll functions. The certificate is permanently linked to the student address at issuance and cannot be moved, sold, or delegated.

3.3 Write Functions

`Function`
`Role`
Description

`addInstitution(addr, name)`
Admin

Registers an institution

`removeInstitution(addr)`
Admin

Deactivates an institution

`transferAdmin(addr)`
Admin

Transfers admin role

`issue(student, name, title, fileHash)`
Institution

Issues a soulbound certificate

`revoke(id)`
Issuing Institution

Revokes a certificate

3.4 Read Functions

`Function`
Description

`getCert(id)`
`Returns full certificate struct by ID`
certsOf(student)

`Returns array of certificate IDs for a student`
issuedBy(institution)

`Returns array of IDs issued by an institution`
totalCerts()

Global certificate count

institutionCount() / institutionAt(i)

Enumerate registered institutions

4. Security Properties

4.1 File Privacy

Only a hash or IPFS CID is stored on-chain — never raw personal data. The actual document lives off-chain (encrypted storage recommended).

4.2 Issuer-Only Revocation

require(c.issuer == msg.sender) ensures only the institution that issued a certificate can revoke it — not the admin, not the student.

4.3 No Upgrade Path

The contract has no proxy, no selfdestruct, and no delegatecall, making its behaviour permanently verifiable.

5. Events

`Event`
`Parameters`
InstitutionAdded

institution address, name

InstitutionRemoved

institution address

Issued

id, issuer, student

Revoked

id, issuer

6. Deployment

Parameter
Value

Chain
BOT Chain (Chain ID 968)

RPC
https://rpc.botchain.ai

Explorer
https://scan.botchain.ai

Admin at deployment
msg.sender (deployer address)

7. Limitations & Roadmap

- Not audited. Use on testnet before production deployment.
- File content is off-chain — loss of the original file is unrecoverable without off-chain backup.
- No batch issuance in v1 — each certificate is a separate transaction.

Milestone
Description

v1.1

Batch certificate issuance (gas savings for large cohorts)

v1.2

ERC-5114 soulbound standard compatibility

v1.3

IPFS CID validation on-chain

Milestone
Description

v2.0

ZK-proof based private verification